

A.2 Forces and momentum

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **27 marks**

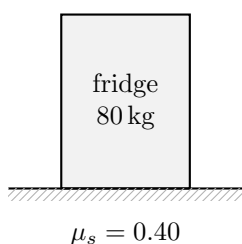
Take $g = 9.81 \text{ m s}^{-2}$. Show your working.

Question 1 — The stubborn fridge

(7 marks)

A flat's 80 kg fridge sits on a floor with coefficient of static friction $\mu_s = 0.40$.

- (a) Draw or list the forces acting on the stationary fridge, and state the value of the normal force. [2]
- (b) Calculate the maximum static friction force available. [2]
- (c) A flatmate pushes horizontally with 250 N and the fridge does not move. State the size of the friction force acting at that moment, and explain why it is *not* the value you calculated in (b). [3]



draw your free-body diagram here

Question 2 — The skate-park catch

(12 marks)

A 55 kg student stands at rest on a skateboard (total mass with board 60 kg, wheels frictionless). A friend throws her a 5.0 kg medicine ball moving horizontally at 6.0 m s^{-1} , and she catches it.

- (a) State the law of conservation of momentum, including the condition under which it holds. [2]
- (b) Calculate the velocity of student, board and ball just after the catch. [3]
- (c) By comparing kinetic energies before and after, determine whether this catch is an elastic or inelastic interaction. [3]
- (d) State where the “missing” kinetic energy has gone. [1]
- (e) Calculate the magnitude of the impulse the ball exerts on the student during the catch. [3]

Question 3 — The falling conker

(4 marks)

A conker falls from a high branch and reaches terminal velocity before landing.

- (a) State the acceleration of the conker and the resultant force on it at terminal velocity. [2]
- (b) Describe how the acceleration changes between release and reaching terminal velocity, and why. [2]

Question 4 — The tug-of-war paradox

(4 marks)

In a tug-of-war, Team A pulls on the rope, and by Newton's third law the rope pulls back on Team A with an equal and opposite force. A puzzled spectator concludes that "all the forces cancel, so nobody can ever win".

- (a) Explain the spectator's error, referring to which body each force of a third-law pair acts on.
[2]
- (b) State what actually decides which team accelerates backwards. [2]

Answers

Question 1 — The stubborn fridge

- (a) Weight mg downwards and the normal force N upwards. Nothing pushes it yet, so there are no horizontal forces at this stage. $N = mg = 80 \times 9.81 = 785 \text{ N}$.
- (b) $f_{\max} = \mu_s N = 0.40 \times 785 = 314 \text{ N}$.
- (c) Exactly 250 N . Static friction is a *responsive* force: it matches whatever is applied, up to its maximum. It only reaches $\mu_s N$ at the point of slipping, and here $250 \text{ N} < 314 \text{ N}$, so the fridge stays put.

Question 2 — The skate-park catch

- (a) The total momentum of a system remains constant provided no resultant external force acts on it.
- (b) $p = 5.0 \times 6.0 = 30 \text{ kg m s}^{-1}$. After the catch the moving mass is $60 + 5.0 = 65 \text{ kg}$, so $v = 30/65 = 0.46 \text{ m s}^{-1}$.
- (c) Before: $E_k = \frac{1}{2}(5.0)(6.0)^2 = 90 \text{ J}$. After: $E_k = \frac{1}{2}(65)(0.46)^2 = 6.9 \text{ J}$. Kinetic energy is not conserved, so the catch is *inelastic* (perfectly inelastic: they move off together). Momentum is still conserved.
- (d) Into internal energy: deformation of ball and body, sound, a slight temperature rise.
- (e) Impulse on the ball $= \Delta p_{\text{ball}} = 5.0(0.46 - 6.0) = -28 \text{ N s}$, so by Newton's third law the student and board receive $+28 \text{ N s}$. (Check on the student and board alone: $60 \times 0.46 = 28 \text{ N s}$.)

Question 3 — The falling conker

- (a) $a = 0$ and the resultant force is zero: drag has grown to balance the weight exactly.
- (b) It starts at g and decreases smoothly to zero. Drag grows with speed, so the resultant force (and hence the acceleration) shrinks as the conker speeds up.

Question 4 — The tug-of-war paradox

- (a) The two forces of a third-law pair act on *different* bodies, one on the rope and one on the team. Forces on different bodies can never cancel each other; only forces on the *same* body can.
- (b) The resultant of the external forces on each team, in practice the friction between their feet and the ground.